

5.1.3 Deep Learning (DL)

ڈیپ لرننگ مشین لرننگ کا وہ طریقہ ہے جس میں کئی پوشیدہ تہوں والے نیورل نیٹ ورک استعمال ہوتے ہیں۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 04 of 15

5.1.3 Deep Learning (DL)

LEARN IT

Deep learning is a machine-learning method in which computers use **multi-layer neural networks** to learn **complex patterns** from data.

The word **“deep”** is used because data passes through **several hidden layers** before the system produces the final output. Each layer learns a **different level of detail**:

- **Early layers** learn simple features — edges, shapes, colours.
- **Deeper layers** combine those simple features to understand complex patterns — a whole face, or a specific car model (a Cultus versus a Corolla).

A network with only one or two hidden layers is a **shallow** neural network. Deep learning is always “deep”: it needs **huge amounts of data** and **processing power**.

Neural network vs deep learning

	Neural network (the model)	Deep learning (the method)
What it is	A system/model that has layers, nodes and connections to help it learn.	A specific method that trains neural networks containing many hidden layers .
Depth	Can be simple (shallow) with just one or two hidden layers.	Always “deep,” requiring huge amounts of data and processing power to learn complex patterns.
Role	The architecture (the “machine”).	The way we train a deep version of that machine.

WORKED EXAMPLE

Real-world example from the source: photo gallery search.

A modern gallery app on a smartphone uses deep learning to recognise faces, objects or places in thousands of photos.

Step 1. During training (done by the company, using millions of images) early layers learned edges and colours; deeper layers learned "wheel", "petal", "minaret".

Step 2. The trained network has already scanned your phone's photos and tagged them.

Step 3. You type "car" or "flower" in the search bar. The app instantly shows related photos because its deep neural network has already learned visual patterns from millions of training images.

A student in Hyderabad can type "Ajrak" or "cricket" and jump to those pictures without scrolling for ten minutes.

REMEMBER IT

Deep = many hidden layers. Shallow NN: 1-2 hidden layers. Deep NN: many hidden layers, huge data, strong computers. Simple features first, complex meaning later.

COMMON MISTAKE

Mistake: "Deep learning and neural networks are two names for the same thing."

Fix: A neural network is the **model** (layers + nodes + connections). Deep learning is the **method** of training networks that have **many** hidden layers. A small NN with one hidden layer is not deep learning.

اردو میں سمجھیں

ڈیپ لرننگ مشین لرننگ کا وہ طریقہ ہے جس میں کئی پوشیدہ تہوں والے نیورل نیٹ ورک استعمال ہوتے ہیں۔ "ڈیپ" اس لیے کہ ڈیٹا کئی تہوں سے گزر کر پیچیدہ پیٹرن سیکھتا ہے: پہلی تہیں کنارے اور رنگ سیکھتی ہیں، گہری تہیں پورا چہرہ یا گاڑی کی قسم پہچانتی ہیں۔ یہ طریقہ بہت سے ڈیٹا اور طاقتور کمپیوٹنگ کا طالب ہے۔

CHECK YOURSELF

1. Why is the method called “deep” learning?
2. What do early layers learn, and what do deeper layers learn?
3. Can a neural network with one hidden layer be called deep learning?

Answers: 1. Because data passes through several hidden layers. 2. Early: simple features (edges, shapes, colours). Deeper: complex patterns (a whole face, a car model). 3. No — that is a shallow network; DL always uses many hidden layers.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Deep learning uses:

- a. Many-layer neural nets
- b. One if-statement
- c. Only Access macros
- d. Only OSI

MCQ 2. "Deep" refers to:

- a. Many hidden layers
- b. Deep sea cables only
- c. Deep stacks in Python only
- d. Deep queues

MCQ 3. DL shines at:

- a. Images, speech, complex patterns
- b. Two-row Excel averages
- c. Primary keys
- d. K-maps of 2 vars

MCQ 4. A cost of DL is:

- a. Lots of data and compute
- b. No data needed
- c. No energy use
- d. Perfect fairness always

MCQ 5. DL is a subset of:

- a. NN / ML / AI
- b. HCI only
- c. SQL only
- d. SDLC only

MCQ 6. Medical imaging in the book is a:

- a. DL application
- b. A foreign key
- c. A pie of NAND
- d. A sprint

Extra short questions

1. Define deep learning.
2. How is it different from a shallow net?
3. Two applications.
4. One Pakistani constraint.

Extra long question

1. Compare ML vs NN vs DL in a 6-row table (definition, depth, data need, example, limit). 10 marks.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. ML using multi-layer neural networks to learn complex patterns.

2. More layers can build higher-level features (edges → shapes → faces).
3. X-ray analysis; voice assistants; translation.
4. GPUs/data/labels are expensive; start with smaller models.

Long

1. Clear nesting. Example row: spam vs digit net vs speech net. Limits: data, bias, black box, energy.

Next lecture: 5.1.4 Components of Neural Networks